

Solar System Exploration

Join us as we explore our solar system.

Planets

Dwarf Planets

10 THINGS ABOUT OUR SOLAR SYSTEM

1. Many Worlds

Our solar system has eight planets, and five dwarf planets.

2. Small Worlds, Too

About 1.4 million asteroids, and about 4,000 comets are in our solar system.

3. Lots of Moons

Our solar system has hundreds of moons.

4. Meet Me in the Milky Way

Our solar system is in one of the Milky Way galaxy's spiral arms called the Orion Spur.

Solar System Facts

Our solar system includes the Sun, eight planets, five officially named dwarf planets, hundreds of moons, and thousands of asteroids and comets.

Our solar system is located in the Milky Way, a barred spiral galaxy with two major arms, and two minor arms. Our Sun is in a small, partial arm of the Milky Way called the Orion Arm, or Orion Spur, between the Sagittarius and Perseus arms. Our solar system orbits the center of the galaxy at about 515,000 mph (828,000 kph). It takes about 230 million years to complete one orbit around the galactic center.

What's Up: July 2026 Skywatching Tips from NASA

A predawn Moon-and-planets meetup, a returning comet, a great chance to see the Milky Way, and Saturn's rings at a...

NASA's Psyche Mission Aces Mars Flyby, Targets Metal-Rich Asteroid

The spacecraft successfully maneuvered around Mars on May 15, speeding it up and setting its course for the main asteroid belt, where it will study the asteroid Psyche in 2029. The flyby also served as a training run, during which Psyche calibrated its instruments ahead of its asteroid encounter, while capturing unique views of the Red Planet.

Mars Gives a Boost to the Psyche Mission

Watch the video to learn about the Psyche spacecraft, its mission, its target — a compelling metal-rich asteroid, also named Psyche — and why the Mars gravity assist was crucial to this journey.

View the Psyche Flyby of Mars, With NASA's 'Eyes on the Solar System'

Use NASA's 3D interactive app to visualize the Psyche spacecraft during its journey. Relive the event, the May 15 gravity assist, when Psyche drew on Mars' gravitational pull to speed up and steer the spacecraft toward its destination in the asteroid belt. Or see where Psyche is right now, beyond Mars and en route to explore the intriguing metal-rich asteroid, also named Psyche, in 2029.

Eyes on the Solar System

NASA's DART Mission Changed Orbit of Asteroid Didymos Around Sun

New research reveals that when NASA's DART (Double Asteroid Redirection Test) spacecraft intentionally impacted the asteroid moonlet Dimorphos in September...

Featured Missions

NEO Surveyor

NEO Surveyor will hunt asteroids and comets that are potential hazards to Earth. Launch is no earlier than September 2027.

Europa Clipper

Europa Clipper launched Oct. 14, 2024, to conduct a detailed investigation of Jupiter's moon Europa.

Psyche

In August 2029, Psyche will arrive at a metal-rich asteroid with the same name.

VERITAS

VERITAS and DAVINCI will be the first NASA spacecraft to explore Venus since the 1990s. VERITAS will launch no earlier than 2031.

Featured Articles

10 Things: Movie Night With NASA

10 Things: Going Interstellar

What's That Space Rock?

10 Things: Our Solar System's Most Marvelous Moons

Resources

Explore curated collections of resources, including activities that can be done at home, as well as videos, animations, handouts, and online interactives.

Asteroids and Comets

Lucy Mission

OSIRIS-REx

Ocean Worlds

View Interstellar Comet 3I/ATLAS Through NASA's Multiple Lenses

Since the interstellar comet was discovered July 1, a dozen NASA assets have captured and processed imagery of the icy interloper, and several others will have opportunities to capture more images as the comet continues to pass through our solar system. Its path runs far beyond Earth; at its closest — on or around Friday, Dec. 19 — comet 3I/ATLAS will be about 170 million miles away, or nearly twice the distance from Earth to the Sun. NASA spacecraft will continue to observe the comet as it makes its journey through the solar system, passing the orbit of Jupiter in spring 2026.

Solar System News

Waxing Gibbous Moon

NASA's Dragonfly Clears Key Tests as Titan Rotorcraft Takes Shape

Curiosity Sees Martian Sulfur Up Close

NASA's New Horizons Spacecraft Wakes from Hibernation in Good Health

NASA's COFFIES Science Center Makes Breakthrough on Solar Enigma

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